

Informed Search: Greedy, A-star, and Heuristic Design

Using problem structure without sacrificing justified guarantees

Learning objectives

By the end of this lesson, you should be able to:

- 01 — Define heuristic, admissibility, consistency, and dominance.
- 02 — Trace Greedy Best-First Search and A-star using g , h , and f .
- 03 — Sketch A-star's optimality argument under admissible or consistent h .
- 04 — Derive heuristics from relaxed problems.
- 05 — Verify Manhattan-distance consistency on four-neighbor grids.
- 06 — Implement deterministic A-star with correct cost updates.

A heuristic injects domain knowledge into ordering

A heuristic $h(n)$ estimates the cheapest remaining cost from node n to a goal.

Estimate

Useful estimates reduce irrelevant expansion while respecting the model's cost units.

- h uses state information unavailable to uninformed ordering.
- h is not the cost already paid.
- A heuristic can be informative, weak, or unsafe.

Greedy and A-star use different priority functions

$$\text{Greedy: } f(n) = h(n) \qquad \text{A-star: } f(n) = g(n) + h(n)$$

- $g(n)$ is exact accumulated cost.
- $h(n)$ estimates remaining optimal cost $h^*(n)$.
- Greedy ignores paid cost.
- A-star balances paid and estimated cost.

ENGINEERING IMPLICATION

Priority and duplicate-update logic must use the same numeric cost model.

Greedy pursues promise; A-star preserves evidence

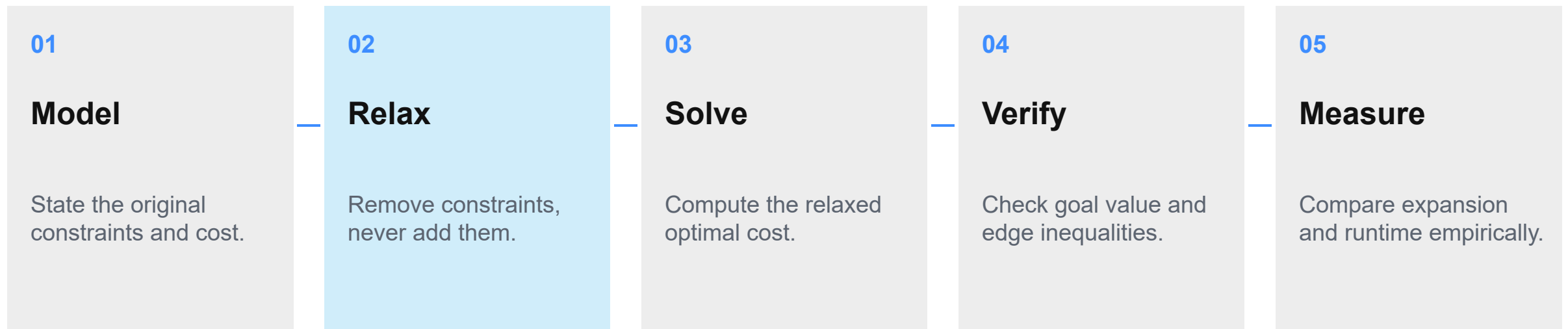
Greedy Best-First

- Priority is h only.
- Can reach a goal quickly.
- Can follow a deceptive heuristic.
- Not generally optimal.

A-star

- Priority is $g+h$.
- Reduces to UCS when $h=0$.
- Optimal under stated h conditions.
- Needs cheaper-path updates.

Design a heuristic through a relaxed problem



A relaxed optimum is a principled lower bound on the original optimum.

A-star extends the UCS priority queue

```
function A_STAR(problem, h):
    push(root, priority=h(start))
    best_g = {start: 0}
    while frontier not empty:
        node = pop_min_f()
        if node.g != best_g[node.state]:
            continue
        if goal(node.state):
            return path(node)
        for child in expand(node):
            if child.g < best_g.get(child.state, infinity):
                best_g[child.state] = child.g
                push(child, child.g + h(child.state))
    return failure
```

Critical details

- Store best g by state.
- Skip stale queue entries.
- Use $h(\text{goal})=0$.
- Break ties deterministically.

Worked A-star trace

SCENARIO

S reaches A with $g=2, h=3$ and B with $g=1, h=5$; A reaches G with cost 3.

- Insert A with $f=5$ and B with $f=6$.
- Remove A because f is smaller.
- Generate G with $g=5, h=0, f=5$.
- Remove G before B and return S-A-G.

A-star returns cost 5; the frontier still contains B with $f=6$.

Heuristic properties answer different questions

Admissible

Node lower bound

- $0 \leq h(n) \leq h^*(n)$.
- Never overestimates.
- Supports A-star tree-search proof.
- $h(\text{goal})=0$ follows.

Consistent

Edge lower bound

- $h(n) \leq c(n, n') + h(n')$.
- Triangle inequality.
- Implies admissibility.
- f does not decrease along paths.

Dominant

Information comparison

- $h_2(n) \geq h_1(n)$.
- Both remain admissible.
- h_2 is at least as informed.
- May reduce expansion.

Checkpoint: test consistency

CHECKPOINT

For an edge A to B with cost 2, $h(A)=5$ and $h(B)=2$. Is the heuristic consistent on this edge?

01

Write the consistency inequality.

02

Substitute the three values.

03

State whether admissibility alone can be inferred.

Manhattan and Euclidean distance encode relaxations

Manhattan distance

- $|dx| + |dy|$.
- Exact without obstacles on four-neighbor unit grids.
- Consistent for unit orthogonal moves.
- Can overestimate if diagonal moves cost 1.

Euclidean distance

- $\sqrt{dx^2 + dy^2}$.
- Straight-line relaxed path.
- Fits continuous or geometric motion.
- Often weaker on four-neighbor grids.

A-star's lower-bound argument protects the optimum

For optimal path node n : $f(n)=g(n)+h(n) \leq g(n)+h^*(n)=C^*$

- C^* is optimal solution cost.
- Admissibility supplies $h(n) \leq h^*(n)$.
- A suboptimal goal G has $f(G)=g(G)>C^*$.
- Minimum- f removal selects an optimal-path node first.

ENGINEERING IMPLICATION

A suboptimal goal cannot be removed while an optimal-path frontier node remains.

Manhattan consistency follows from one move

SCENARIO

A legal grid move changes exactly one coordinate by one and costs 1.

- Let h be Manhattan distance to the goal.
- One move changes h by at most 1.
- Therefore $h(s) \leq 1 + h(s')$.
- The edge consistency inequality holds.

Manhattan distance is consistent and therefore admissible for this movement model.

Heuristic failure modes are model mismatches

Overestimate

Unsafe lower bound

- Can lose optimality.
- Often caused by wrong move costs.
- Goal value may be nonzero.
- Counterexample needed.

Too weak

Little guidance

- Behaves like UCS.
- Optimal but expands more.
- Relaxation removed too much.
- Measure expansion.

Inconsistent

Decreasing f

- Graph search may reopen states.
- Stale entries appear.
- Local edge test detects it.
- Correctness policy must match.

Checkpoint: choose a grid heuristic

CHECKPOINT

A grid permits eight directions; orthogonal moves cost 1 and diagonal moves cost $\sqrt{2}$. Is Manhattan distance admissible?

01

Compare one diagonal displacement.

02

Compute Manhattan and true relaxed cost.

03

Name a better geometric heuristic.

Heuristics require both proof and measurement

Validate mathematical safety separately from empirical usefulness.

- Unit-test $h(\text{goal})=0$ and sampled edge consistency.
- Compare solution cost against UCS on small instances.
- Report expansions, maximum frontier, and wall-clock time.

Evidence

A proof supports optimality;
benchmarks support efficiency
claims.

Key takeaways

- 01 — Greedy orders by h ; A-star orders by $g+h$.
- 02 — Admissibility is a global lower bound; consistency is an edge inequality.
- 03 — Relaxed problems provide principled heuristics.
- 04 — Correct A-star needs best- g updates and deterministic queue behavior.

Next: Week 3 replaces a passive environment with an optimizing opponent and introduces minimax.

Check your understanding

Q1

Trace A-star on a five-node weighted graph.

Q2

Give an admissible but inconsistent heuristic.

Q3

Prove Manhattan consistency for one grid edge.

Q4

Explain heuristic dominance.

Q5

Design tests for an A-star implementation.